Chapter 13

Representation Theory

Representation theory studies groups via their actions on vector spaces, i.e., homomorphisms $G \to GL(V)$. This converts abstract group theory into linear algebra, where we have powerful tools (eigenvalues, traces, diagonalization). We work over $k = \mathbb{C}$ throughout (or more generally, any algebraically closed field of characteristic 0).

13.1 Representations and Subrepresentations

Definition 13.1.1: Representation

A representation of a group G is a vector space V (usually finite-dimensional over $\mathbb{C}$) together with a group homomorphism $\rho : G \to GL(V)$. Equivalently, a G-action on V by linear operators: $G \times V \to V$, $(g, v) \mapsto g \cdot v$, with each $g : V \to V$ linear.

Definition 13.1.2: Subrepresentation and irreducibility

- A subrepresentation is a subspace $W \subseteq V$ that is G-invariant: $gW = W$ for all $g \in G$.
- A representation V is irreducible if it has no nontrivial subrepresentations (i.e., the only G-invariant subspaces are $\{0\}$ and V itself).

Example 13.1.1 (Representations of cyclic groups)

If $G = \mathbb{Z}/n$, a representation is a vector space V with a linear operator $\varphi = \rho(1) : V \to V$ satisfying $\varphi^n = \mathrm{id}_V$. Since the minimal polynomial of φ divides $x^n - 1 = \prod_{k=0}^{n-1}(x - \lambda_k)$ where $\lambda_k = e^{2\pi i k/n}$, and this polynomial has distinct roots, φ is diagonalizable. The eigenspaces $V_{\lambda_k} = \{v : \varphi(v) = \lambda_k v\}$ are subrepresentations, and $V = \bigoplus_k V_{\lambda_k}$.
Each 1-dimensional subspace $\mathbb{C}v$ with $\varphi(v) = \lambda_k v$ is an irreducible representation, corresponding to the homomorphism $\mathbb{Z}/n \to \mathbb{C}^* = GL_1(\mathbb{C})$ sending $1 \mapsto \lambda_k = e^{2\pi i k/n}$. There are exactly n such irreducible representations, and every representation of $\mathbb{Z}/n$ decomposes as a direct sum of copies of these.

Proposition 13.1.1 Representations of finite abelian groups

If $G \cong \mathbb{Z}/m_1 \times \cdots \times \mathbb{Z}/m_r$ is a finite abelian group and V is a representation of G over $\mathbb{C}$, then V splits into a direct sum of 1-dimensional (irreducible) subrepresentations.

Proof: The G-action is equivalent to commuting operators $\varphi_1, \ldots, \varphi_r : V \to V$ with $\varphi_i^{m_i} = \mathrm{id}$. Each φ_i is diagonalizable (as above). Commuting diagonalizable operators are simultaneously diagonalizable: the eigenspaces of φ_1 are invariant under all φ_j (since they commute), and the restriction of φ_j to an eigenspace of φ_1 is again of finite order, hence diagonalizable. Proceed by induction on r. The simultaneous eigenspaces are 1-dimensional subrepresentations. □

Definition 13.1.3: Dual group

For a finite abelian group G, the dual group is $\hat{G} = \mathrm{Hom}(G, \mathbb{C}^*)$, the group of all 1-dimensional representations (characters in the abelian sense) under pointwise multiplication. For $G = \mathbb{Z}/n$, $\hat{G} \cong \mathbb{Z}/n$ (but not canonically). More generally, $\hat{G} \cong G$ for any finite abelian group (non-canonically). This completes the classification of complex representations of finite abelian groups.

13.1.1 Homomorphisms and Constructions

Definition 13.1.4: Homomorphism of representations

Given representations V, W of G, a homomorphism (or G-equivariant map, or intertwining operator) is a linear map $\varphi : V \to W$ satisfying $\varphi(g \cdot v) = g \cdot \varphi(v)$ for all $g \in G$, $v \in V$. Equivalently, $g \circ \varphi = \varphi \circ g$ as operators. We write $\mathrm{Hom}_G(V, W)$ for the space of such maps.

Note:-

If $\varphi \in \mathrm{Hom}_G(V, W)$, then $\mathrm{Ker}(\varphi)$ and $\mathrm{Im}(\varphi)$ are subrepresentations of V and W respectively (since $v \in \mathrm{Ker}(\varphi) \implies \varphi(gv) = g\varphi(v) = 0$, so $gv \in \mathrm{Ker}(\varphi)$).

Proposition 13.1.2 Constructions with representations

Given representations V, W of G:

(i) $V \oplus W$ is a representation: $g \cdot (v, w) = (gv, gw)$.

(ii) $V \otimes W$ is a representation: $g \cdot (v \otimes w) = gv \otimes gw$, extended by linearity.

(iii) The dual $V^* = \mathrm{Hom}(V, \mathbb{C})$ is a representation: $(g \cdot \ell)(v) = \ell(g^{-1}v)$. This is the contragredient representation.

(iv) $\mathrm{Hom}(V, W)$ is a representation: $(g \cdot \varphi)(v) = g \cdot \varphi(g^{-1}v)$, i.e., $g \cdot \varphi = g \circ \varphi \circ g^{-1}$. Note $\mathrm{Hom}(V, W)^G = \mathrm{Hom}_G(V, W)$ (the G-invariant maps are exactly the G-equivariant ones).

(v) Under the isomorphism $\mathrm{Hom}(V, W) \cong V^* \otimes W$, the G-action is $g \cdot (\ell \otimes w) = (g \cdot \ell) \otimes (g \cdot w)$, which is consistent with the above.

(vi) If $W \subseteq V$ is a subrepresentation, then V/W is a representation: $g \cdot (v + W) = gv + W$.

13.2 Maschke's Theorem and Complete Reducibility

Lemma 13.2.1 Existence of G-invariant inner product

If V is a $\mathbb{C}$-representation of a finite group G, there exists a positive-definite Hermitian inner product H on V that is G-invariant: $H(gv, gw) = H(v, w)$ for all $g \in G$, $v, w \in V$. In other words, all operators $g : V \to V$ are unitary with respect to H.

Proof: Start with any Hermitian inner product H_0 on V, and average over G:

$$H(v, w) = \frac{1}{|G|} \sum_{g \in G} H_0(gv, gw).$$

Then H is still Hermitian and positive-definite (a positive combination of positive-definite forms), and G-invariant: $H(hv, hw) = \frac{1}{|G|}\sum_g H_0(ghv, ghw) = \frac{1}{|G|}\sum_{g'} H_0(g'v, g'w) = H(v, w)$ (reindexing $g' = gh$). $\square$

Theorem 13.2.1 Maschke's Theorem

Let V be a representation of a finite group G over $\mathbb{C}$ (or any field k with $\mathrm{char}(k) \nmid |G|$), and let $W \subseteq V$ be a subrepresentation. Then there exists a complementary subrepresentation $U \subseteq V$ with $V = W \oplus U$.

Proof 1 (via invariant inner product): Equip V with a G-invariant Hermitian inner product H (by the lemma). Since each g is unitary, $g(W^\perp) = W^\perp$ (if $w \in W$ and $v \in W^\perp$, then $H(gv, w) = H(v, g^{-1}w) = 0$ since $g^{-1}w \in W$). So $U = W^\perp$ is a G-invariant complement. $\square$

Proof 2 (via averaging projection): Choose any complementary subspace U_0 with $V = W \oplus U_0$ (not necessarily G-invariant). Let $\pi_0 : V \to W$ be the projection onto W with kernel U_0 ($\pi_0|_W = \mathrm{id}$, $\pi_0|_{U_0} = 0$). Average:

$$\pi(v) = \frac{1}{|G|}\sum_{g\in G} g\,\pi_0(g^{-1}v).$$

Then $\pi : V \to W$ is G-equivariant ($g\pi g^{-1} = \pi$), and $\pi|_W = \mathrm{id}$ (since $g\pi_0(g^{-1}w) = g(g^{-1}w) = w$ for $w \in W$, as $g^{-1}w \in W$). So π is a surjection onto V^G, and $U = \mathrm{Ker}(\pi)$ is a G-invariant complement. $\square$

Note:-

Maschke's theorem fails when $\mathrm{char}(k)$ divides $|G|$ (modular representation theory), and also fails for infinite groups in general: the representation of $G = \mathbb{Z}$ (or $\mathbb{R}$) on $\mathbb{C}^2$ via $t \mapsto \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ has the invariant subspace $\mathbb{C} \times \{0\}$ but no complementary invariant subspace.

Corollary 13.2.1 Complete reducibility

Any finite-dimensional representation of a finite group G (over $\mathbb{C}$) decomposes as a direct sum of irreducible representations.

13.3 Schur's Lemma

Theorem 13.3.1 Schur's Lemma

Let V, W be irreducible representations of G, and $\varphi : V \to W$ a homomorphism of representations. Then:

(i) Either $\varphi = 0$ or φ is an isomorphism.

(ii) Over $k = \mathbb{C}$: if V is irreducible and $\varphi : V \to V$ is a homomorphism of representations (i.e., $\varphi \in \mathrm{Hom}_G(V, V)$), then $\varphi = \lambda \cdot \mathrm{id}$ for some $\lambda \in \mathbb{C}$.

Proof: (i) $\mathrm{Ker}(\varphi)$ is a subrepresentation of V. Since V is irreducible, either $\mathrm{Ker}(\varphi) = V$ ($\varphi = 0$) or $\mathrm{Ker}(\varphi) = \{0\}$ (φ injective). Similarly, $\mathrm{Im}(\varphi)$ is a subrepresentation of W, so either $\mathrm{Im}(\varphi) = \{0\}$ ($\varphi = 0$) or $\mathrm{Im}(\varphi) = W$ (φ surjective).

(ii) Over $\mathbb{C}$, φ has an eigenvalue λ. Then $\varphi - \lambda \cdot \mathrm{id}$ is also a G-homomorphism $V \to V$ with nontrivial kernel. By (i), $\varphi - \lambda \cdot \mathrm{id} = 0$. $\square$

Example 13.3.1 (Center acts by scalars)

Let V be an irreducible representation of G, and $h \in Z(G)$. Since h commutes with all $g \in G$, the map $\rho(h) : V \to V$ commutes with all $\rho(g)$, i.e., $\rho(h) \in \mathrm{Hom}_G(V, V)$. By Schur, $\rho(h) = \lambda_h \cdot \mathrm{id}$ for some

$\lambda_h \in \mathbb{C}$. So every central element acts as a scalar on every irreducible representation. In particular, irreducible representations of abelian groups are 1-dimensional (giving another proof of the earlier result).

13.3.1 Irreducible Representations of S_3

Example 13.3.2 (The three irreducible representations of S_3)

- The trivial representation $U = \mathbb{C}$: $\sigma \cdot z = z$ for all $\sigma \in S_3$.
- The alternating (sign) representation $U' = \mathbb{C}$: $\sigma \cdot z = \mathrm{sgn}(\sigma)z$.
- The standard representation V: start with the permutation representation $\mathbb{C}^3$ (basis e_1, e_2, e_3, with $\sigma \cdot e_i = e_{\sigma(i)}$). This has an invariant subspace $\mathrm{span}(e_1 + e_2 + e_3) \cong U$. The complement $V = \{(z_1, z_2, z_3) \in \mathbb{C}^3 : z_1 + z_2 + z_3 = 0\}$ has $\dim V = 2$ and is irreducible.

Irreducibility of V: Let $\tau = (123)$ and $\sigma = (12)$. Then τ acts on V with eigenvalues $\lambda = e^{2\pi i/3}$ and $\lambda^2 = e^{4\pi i/3}$ (since $\tau^3 = \mathrm{id}$ and $\tau|_V \neq \mathrm{id}$). If v is a λ-eigenvector, then $\sigma(v)$ is a λ^2-eigenvector (since $\tau(\sigma v) = \sigma(\tau^{-1} v) = \sigma(\lambda^2 v) = \lambda^2 \sigma v$, using $\sigma\tau\sigma^{-1} = \tau^{-1}$). Since $\lambda \neq \lambda^2$, v and $\sigma(v)$ are linearly independent, so they span V. Any invariant subspace of V must contain an eigenvector of τ (since τ is diagonalizable on V), hence contains v or $\sigma(v)$, and then contains $\mathrm{span}(v, \sigma(v)) = V$. So V is irreducible. $\square$

Proposition 13.3.1 These are all irreducibles of S_3

The representations U, U', V are the only irreducible representations of S_3 over $\mathbb{C}$, and any representation W of S_3 decomposes as $W \cong U^{\oplus a} \oplus U'^{\oplus b} \oplus V^{\oplus c}$ for unique $a, b, c \geqslant 0$.

Note:-

We will prove uniqueness (and a method to find a, b, c) using characters below. The key dimension check: $\dim U^2 + \dim U'^2 + \dim V^2 = 1 + 1 + 4 = 6 = |S_3|$, which we will show is a necessary condition.

13.4 Characters

Definition 13.4.1: Character

The character of a representation V is the function $\chi_V : G \to \mathbb{C}$ defined by $\chi_V(g) = \mathrm{Tr}(\rho(g))$, the trace of the linear operator $g : V \to V$.

Note:-

Since trace is conjugation-invariant ($\mathrm{Tr}(ABA^{-1}) = \mathrm{Tr}(B)$), the character only depends on the conjugacy class of g. That is, χ_V is a class function.

Definition 13.4.2: Class function

A class function is a function $f : G \to \mathbb{C}$ satisfying $f(ghg^{-1}) = f(h)$ for all $g, h \in G$.

Proposition 13.4.1 Properties of characters

Given representations V, W of G:

(i) $\chi_{V \oplus W}(g) = \chi_V(g) + \chi_W(g)$.

(ii) $\chi_{V\otimes W}(g) = \chi_V(g)\cdot\chi_W(g)$.

(iii) $\chi_{V^*}(g) = \overline{\chi_V(g)}$ (since g acts by ${}^t(g^{-1})$, and eigenvalues of g are roots of unity, so eigenvalues of g^{-1} are their conjugates).

(iv) $\chi_{\mathrm{Hom}(V,W)}(g) = \overline{\chi_V(g)}\cdot\chi_W(g)$.

(v) $\chi_V(e) = \dim V$.

(vi) For a permutation representation on a set S: $\chi_V(g) = |\{s\in S : g\cdot s = s\}| = |S^g|$.

(vii) $\chi_{\bigwedge^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 - \chi_V(g^2))$.

13.4.1 Symmetric Polynomials (Digression)

The connection between eigenvalues and characters motivates a brief digression. To store information about n complex numbers (the eigenvalues of $g : V \to V$), up to reordering, it suffices to specify the coefficients of the polynomial $\prod(x-\lambda_i)$. These coefficients are the elementary symmetric polynomials $\sigma_k(z_1,\ldots,z_n) = \sum_{i_1<\cdots<i_k} z_{i_1}\cdots z_{i_k}$.

Theorem 13.4.1 Fundamental theorem of symmetric polynomials

The subring of symmetric polynomials $\mathbb{C}[z_1,\ldots,z_n]^{S_n}$ is isomorphic to the polynomial algebra $\mathbb{C}[\sigma_1,\ldots,\sigma_n]$. That is, every symmetric polynomial is uniquely a polynomial expression in the elementary symmetric polynomials.

Note:-

Alternatively, one can use the power sums $\tau_k = \sum z_i^k$. Then $\mathbb{C}[z_1,\ldots,z_n]^{S_n} \cong \mathbb{C}[\tau_1,\ldots,\tau_n]$ as well. In particular, specifying the unordered tuple $\{z_1,\ldots,z_n\}$ is equivalent to specifying $\sum z_i, \sum z_i^2, \ldots, \sum z_i^n$. For representations, this means specifying all eigenvalues of g is equivalent to specifying $\mathrm{Tr}(g), \mathrm{Tr}(g^2), \ldots, \mathrm{Tr}(g^n)$. But since G is a group, $\mathrm{Tr}(g^k)$ is just $\chi_V(g^k)$—which is already determined by χ_V. So the character determines the eigenvalues.

13.5 Character Orthogonality

13.5.1 The Inner Product on Class Functions

Proposition 13.5.1 Averaging projection

If V is a representation of G, the map $\varphi_V = \frac{1}{|G|}\sum_{g\in G} g : V \to V$ is a projection onto $V^G = \{v \in V : gv = v\ \forall g\}$, the space of G-invariant vectors. In particular, $\dim V^G = \mathrm{Tr}(\varphi_V) = \frac{1}{|G|}\sum_{g\in G}\chi_V(g)$.

Note:-

For representations V, W: $\mathrm{Hom}_G(V,W) = \mathrm{Hom}(V,W)^G$, so $\dim \mathrm{Hom}_G(V,W) = \frac{1}{|G|}\sum_{g\in G}\chi_{\mathrm{Hom}(V,W)}(g) = \frac{1}{|G|}\sum_{g\in G}\overline{\chi_V(g)}\chi_W(g)$. If V and W are irreducible, Schur's lemma gives $\dim\mathrm{Hom}_G(V,W) = \begin{cases} 1 & V\cong W \\ 0 & V\not\cong W \end{cases}$.

Definition 13.5.1: Hermitian inner product on class functions

Define $H : \{\text{class functions}\} \times \{\text{class functions}\} \to \mathbb{C}$ by

$$H(\alpha, \beta) = \frac{1}{|G|} \sum_{g \in G} \overline{\alpha(g)} \beta(g).$$

Theorem 13.5.1 Character orthogonality

The characters of the irreducible representations of G are orthonormal with respect to H. That is, if $V_1, \dots, V_k$ are the distinct irreducible representations of G, then

$$H(\chi_{V_i}, \chi_{V_j}) = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_{V_i}(g)} \chi_{V_j}(g) = \delta_{ij}.$$

In particular, the characters of distinct irreducible representations are linearly independent as class functions.

Corollary 13.5.1 Number of irreducibles

The number of irreducible representations of G is at most the number of conjugacy classes of G (since the irreducible characters are linearly independent class functions, and the space of class functions has dimension = number of conjugacy classes). In fact, equality holds: the number of irreducible representations equals the number of conjugacy classes.

Corollary 13.5.2 Character determines representation

Every representation of G is completely determined (up to isomorphism) by its character. If $W \cong \bigoplus V_i^{\oplus a_i}$, then $\chi_W = \sum a_i \chi_{V_i}$, and $a_i = H(\chi_{V_i}, \chi_W)$.

Corollary 13.5.3 Irreducibility criterion

A representation W is irreducible if and only if $H(\chi_W, \chi_W) = 1$. More generally, $H(\chi_W, \chi_W) = \sum a_i^2$ where $W \cong \bigoplus V_i^{\oplus a_i}$, so the inner product counts the "number of irreducible summands" (with multiplicity squared).

13.5.2 The Regular Representation

Definition 13.5.2: Regular representation

The regular representation R of G is the vector space $\mathbb{C}[G]$ with basis $\{e_g\}_{g \in G}$, on which G acts by left multiplication: $g \cdot e_h = e_{gh}$.

Proposition 13.5.2 Decomposition of the regular representation

In the regular representation, each irreducible V_i appears with multiplicity $\dim V_i$: $R \cong \bigoplus_i V_i^{\oplus \dim V_i}$.

Proof: $\chi_R(g) = |\{h \in G : gh = h\}| = \begin{cases} |G| & g = e \\ 0 & g \neq e \end{cases}$ (permutation representation where only e has fixed points). Then $a_i = H(\chi_{V_i}, \chi_R) = \frac{1}{|G|} \sum_g \overline{\chi_{V_i}(g)} \chi_R(g) = \frac{1}{|G|} \overline{\chi_{V_i}(e)} \cdot |G| = \chi_{V_i}(e) = \dim V_i$. □

Corollary 13.5.4 Dimension formula

$\sum_i (\dim V_i)^2 = |G|$, where the sum is over all irreducible representations.

Proof: $\dim R = |G| = \sum_i a_i \dim V_i = \sum_i (\dim V_i)^2$. □

13.6 Character Tables

13.6.1 Character Table of S_3

S_3 has 3 conjugacy classes: $\{e\}$, $\{(12), (13), (23)\}$, $\{(123), (132)\}$. The 3 irreducible representations have dimensions satisfying $d_1^2 + d_2^2 + d_3^2 = 6$, so $d_i \in \{1, 1, 2\}$.

	e	(12)	(123)
	1	3	2
U (trivial)	1	1	1
U' (sign)	1	-1	1
V (standard)	2	0	-1

The character of V is obtained from the permutation representation: $\chi_{U\oplus V}(g) = |S^g|$, giving $\chi_{U\oplus V} = (3, 1, 0)$, so $\chi_V = (2, 0, -1)$. Check: $H(\chi_V, \chi_V) = \frac{1}{6}(4 + 0 + 2) = 1$, confirming irreducibility.

13.6.2 Character Table of S_4

S_4 has 5 conjugacy classes (partitions of 4): $\{e\}$ (1), transpositions (6), double transpositions (3), 3-cycles (8), 4-cycles (6). The dimension formula $\sum d_i^2 = 24$ with 5 irreducibles gives $d_i \in \{1, 1, 2, 3, 3\}$.

Three known irreducibles: U (trivial, dim 1), U' (sign, dim 1), V (standard, dim 3: permutation rep on $\mathbb{C}^4$ minus trivial summand). The tensor product $V' = V \otimes U'$ (twist standard by sign) has $\chi_{V'}(g) = \chi_V(g) \cdot \mathrm{sgn}(g)$ and is irreducible ($H(\chi_{V'}, \chi_{V'}) = 1$) and distinct from V. This gives 4 of the 5; we need one more of dimension 2 (since $1 + 1 + 9 + 9 = 20$, and $24 - 20 = 4 = 2^2$).

For the missing W (dim 2): use $V \otimes V$. We have $\chi_{V\otimes V} = \chi_V^2 = (9, 1, 1, 0, 1)$. Compute inner products with known characters to find $V \otimes V \cong U \oplus V \oplus V' \oplus W$. Then $\chi_W = \chi_{V\otimes V} - \chi_U - \chi_V - \chi_{V'} = (2, 0, 2, -1, 0)$. Check: $H(\chi_W, \chi_W) = \frac{1}{24}(4 + 0 + 12 + 8 + 0) = 1$.

The complete character table:

	e	(12)	$(12)(34)$	(123)	(1234)
	1	6	3	8	6
U	1	1	1	1	1
U'	1	-1	1	1	-1
V	3	1	-1	0	-1
V'	3	-1	-1	0	1
W	2	0	2	-1	0

Note:-

The representation W has $\chi_W((12)(34)) = 2 = \dim W$, so double transpositions act trivially. The normal subgroup $H = \{e, (12)(34), (13)(24), (14)(23)\} \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ acts trivially on W, so W factors through $S_4/H \cong S_3$. In fact, W is the standard representation of S_3 "pulled back" to S_4 via the quotient map.

13.6.3 Character Table of A_4

A_4 has 4 conjugacy classes: $\{e\}$ (1), $\{(12)(34), (13)(24), (14)(23)\}$ (3), $\{(123), (134), (243), (142)\}$ (4), $\{(132), (143), (234), (124)\}$ (4). Note the 3-cycles split into two classes (since A_4 has no odd permutations to conjugate them). The dimension formula $\sum d_i^2 = 12$ gives $d_i \in \{1, 1, 1, 3\}$.

The three 1-dimensional representations correspond to $\mathrm{Hom}(A_4, \mathbb{C}^*) \cong \mathrm{Hom}(A_4/H, \mathbb{C}^*) \cong \widehat{\mathbb{Z}/3} = \{1, \omega, \omega^2\}$ where $\omega = e^{2\pi i/3}$, and the 3-dimensional irreducible is the restriction of the standard representation of S_4.

	e	$(12)(34)$	(123)	(132)
	1	3	4	4
U	1	1	1	1
U'	1	1	ω	ω^2
U''	1	1	ω^2	ω
V	3	-1	0	0

Exercise 13.6.1 Representation theory exercises

(1) Decompose the permutation representation $\mathbb{C}^3$ of S_3 into irreducibles and verify using the character inner product.

(2) Let V be the standard representation of S_3. Decompose $V \otimes V$, $\mathrm{Sym}^2(V)$, and $\bigwedge^2 V$ into irreducibles.

(3) Compute the character table of $\mathbb{Z}/2 \times \mathbb{Z}/2$ (the Klein four-group). How many irreducible representations are there, and what are their dimensions?

(4) Using the character table of S_4, verify that $V \otimes V' \cong U' \oplus W \oplus V \oplus V'$ (where V, V', W are as in the table).

(5) Prove that for any representation V of a finite group G, $\frac{1}{|G|}\sum_{g\in G} \chi_V(g) = \dim V^G$.

Solution

(1) The permutation representation has character $\chi(g) = |S^g|$: $\chi(e) = 3$, $\chi((12)) = 1$, $\chi((123)) = 0$. Compute: $H(\chi_U, \chi) = \frac{1}{6}(3+3+0) = 1$, $H(\chi_{U'}, \chi) = \frac{1}{6}(3-3+0) = 0$, $H(\chi_V, \chi) = \frac{1}{6}(6+0+0) = 1$. So $\mathbb{C}^3 \cong U \oplus V$.
(2) $\chi_{V\otimes V}(g) = \chi_V(g)^2$: values $(4, 0, 1)$. Inner products: $H(\chi_U, \chi_{V\otimes V}) = \frac{1}{6}(4+0+2) = 1$, $H(\chi_{U'}, \chi_{V\otimes V}) = \frac{1}{6}(4+0+2) = 1$, $H(\chi_V, \chi_{V\otimes V}) = \frac{1}{6}(8+0-2) = 1$. So $V \otimes V \cong U \oplus U' \oplus V$.
For $\mathrm{Sym}^2(V)$: $\chi_{\mathrm{Sym}^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 + \chi_V(g^2))$. Values: $\frac{1}{2}(4+2, 0+2, 1+(-1)) = (3,1,0)$. So $\chi_{\mathrm{Sym}^2 V} = \chi_U + \chi_V$, giving $\mathrm{Sym}^2 V \cong U \oplus V$.
For $\bigwedge^2 V$: $\chi_{\bigwedge^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 - \chi_V(g^2))$. Values: $\frac{1}{2}(4-2, 0-2, 1+1) = (1,-1,1) = \chi_{U'}$. So $\bigwedge^2 V \cong U'$. (This makes sense: the determinant map $\bigwedge^2 V \to \mathbb{C}$ of a 2-dimensional rep is the sign character.)
(3) $\mathbb{Z}/2 \times \mathbb{Z}/2$ is abelian with 4 elements, so it has 4 conjugacy classes and 4 one-dimensional irreducibles. The dual group is $\widehat{\mathbb{Z}/2 \times \mathbb{Z}/2} \cong \mathbb{Z}/2\times\mathbb{Z}/2$. Writing elements as $\{(0,0), (1,0), (0,1), (1,1)\}$, the 4 characters are $\chi_{ab}(x,y) = (-1)^{ax+by}$ for $(a,b) \in \{0,1\}^2$.
(4) $\chi_{V\otimes V'} = \chi_V \cdot \chi_{V'}$: values $(9, -1, 1, 0, -1)$. Compute H with each irreducible to find multiplicities. This yields $V \otimes V' \cong U' \oplus W \oplus V \oplus V'$ (verify the character sum matches).
(5) The averaging operator $\varphi = \frac{1}{|G|}\sum_g g$ projects onto V^G. Taking traces: $\mathrm{Tr}(\varphi) = \frac{1}{|G|}\sum_g \mathrm{Tr}(g) = \frac{1}{|G|}\sum_g \chi_V(g)$. Since φ is a projection ($\varphi^2 = \varphi$), its trace equals its rank: $\mathrm{Tr}(\varphi) = \dim \mathrm{Im}(\varphi) = \dim V^G$.

13.7 Characters Form an Orthonormal Basis

We proved that the irreducible characters $\chi_{V_1}, \ldots, \chi_{V_k}$ are orthonormal with respect to H. This shows they are linearly independent, giving $k \leqslant$ (number of conjugacy classes). We now prove they also *span* the space of class functions, completing the proof that k equals the number of conjugacy classes.

13.7.1 The Generalized Projection Operator

The averaging projection $\varphi_V = \frac{1}{|G|}\sum_{g\in G} g$ onto V^G can be generalized. Given any function $\alpha : G \to \mathbb{C}$ and any representation V of G, define

$$\varphi_{\alpha,V} = \frac{1}{|G|}\sum_{g\in G} \alpha(g)\cdot g : V \to V.$$

This is a "weighted average" of the group elements acting on V.

Proposition 13.7.1 Equivariance of φ_α

If $\alpha : G \to \mathbb{C}$ is a class function, then $\varphi_{\alpha,V}$ is G-equivariant, i.e., $\varphi_{\alpha,V} \in \mathrm{Hom}_G(V, V)$.

Proof: We must show $h \cdot \varphi_\alpha(v) = \varphi_\alpha(h \cdot v)$ for all $h \in G$, $v \in V$. For the left side:

$$h \cdot \varphi_\alpha(v) = \frac{1}{|G|}\sum_{g\in G} \alpha(g)\cdot hg(v).$$

Relabeling $k = hg$ (so $g = h^{-1}k$):

$$h \cdot \varphi_\alpha(v) = \frac{1}{|G|}\sum_{k\in G} \alpha(h^{-1}k)\cdot k(v).$$

For the right side:

$$\varphi_\alpha(hv) = \frac{1}{|G|}\sum_{g\in G} \alpha(g)\cdot g(hv) = \frac{1}{|G|}\sum_{g\in G} \alpha(g)\cdot gh(v).$$

Relabeling $k = gh$ (so $g = kh^{-1}$):

$$\varphi_\alpha(hv) = \frac{1}{|G|}\sum_{k\in G} \alpha(kh^{-1})\cdot k(v).$$

These are equal if and only if $\alpha(h^{-1}k) = \alpha(kh^{-1})$ for all $h, k \in G$, which holds precisely when α is a class function (since $kh^{-1} = k(h^{-1}k)k^{-1}$, so $h^{-1}k$ and kh^{-1} are conjugate). $\square$